

Anand Model: Viscoelastoplasticity

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and its Application to Solder Joints

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Constitutive Equations for Hot-Working of Metals

Author: Lallit Anand (1985)

DOI: 10.1016/0749-6419(85)90004-X

One of the foundational papers in thermodynamically consistent viscoplasticity modeling—especially significant in the context of metals subjected to large strains and high temperatures.

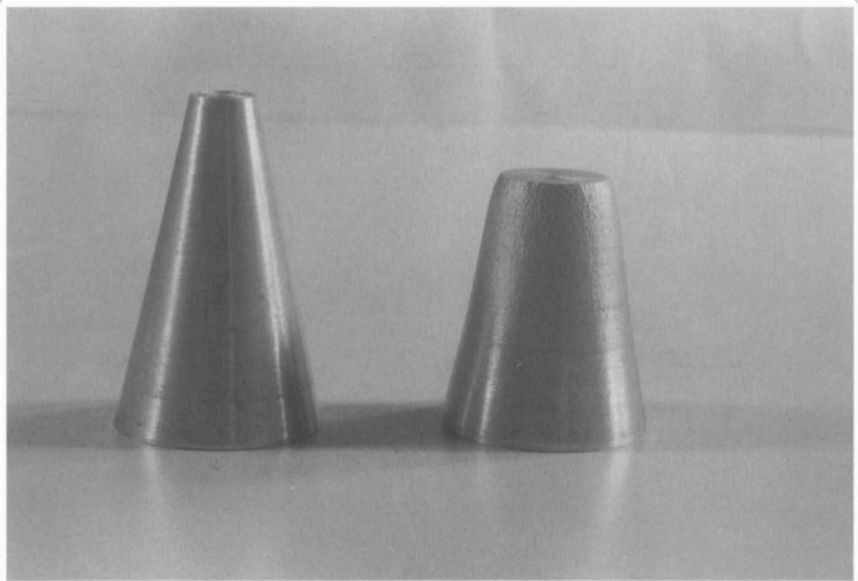


Fig. 25. 1100 aluminum state gradient specimens before and after testing.

CONSTITUTIVE EQUATIONS FOR HOT-WORKING OF METALS

LALLIT ANAND

Massachusetts Institute of Technology

(Communicated by Theodor Lehmann, Ruhr Universität Bochum)

Abstract—Elevated temperature deformation processing—"hot-working," is an important step during the manufacturing of most metal products. Central to any successful analysis of a hot-working process is the use of appropriate rate and temperature-dependent constitutive equations for large, interrupted inelastic deformations, which can faithfully account for strain-hardening, the restoration processes of recovery and recrystallization and strain rate and temperature history effects. In this paper we develop a set of phenomenological, internal variable type constitutive equations describing the elevated temperature deformation of metals. We use a scalar and a symmetric, traceless, second-order tensor as internal variables which, in an average sense, represent an isotropic and an anisotropic resistance to plastic flow offered by the internal state of the material. In this theory, we consider small elastic stretches but large plastic deformations (within the limits of texturing) of isotropic materials. Special cases (within the constitutive framework developed here) which should be suitable for analyzing hot-working processes are indicated.

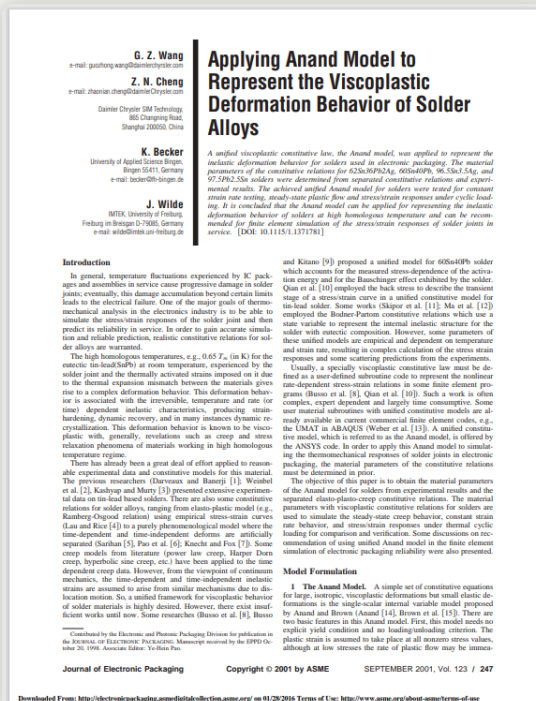
1. INTRODUCTION

Hot-working is an important processing step during the manufacture of approximately more than eighty-five percent of all metal products. The main features of hot-working are that metals are deformed into the desired shapes at temperatures in the range of -0.5 through $-0.9 \theta_m$, where θ_m is the melting temperature in degrees Kelvin, and at strain rates in the range of -10^{-4} through $-10^3/\text{sec}$. It is to be noted that most hot-working processes are more than mere shape-making operations; an important goal of hot-working is to subject the workpiece to appropriate thermo-mechanical processing histories which will produce microstructures that optimize the mechanical properties of the product.

The major quantities of metals and alloys are hot-worked under interrupted non-isothermal conditions. The principles of the physical metallurgy of such deformation processing are now well recognized, e.g., JONAS *et al.* [1969], SELLARS & MCG TEGART [1972], McQUEEN & JONAS [1975], and SELLARS [1978]. During a deformation pass, the stress is found to be a strong function of the strain rate, temperature, and the defect and microstructural state of the material. The strain-hardening produced by the deformation tends to be counteracted by dynamic recovery processes. These recovery processes result in a rearrangement and annihilation of dislocations in such a manner that as the strain in a pass increases, the dislocations tend to arrange themselves into sub-grain walls. In some metals and alloys (especially those with a high stacking fault energy, e.g., Al, α -Fe and other ferritic alloys) dynamic recovery can balance strain-hardening and an apparent steady state stress level can be achieved and maintained to large strains before fracture occurs. In other metals and alloys in which recovery is less rapid (especially those metals with low stacking fault energies, e.g., Ni, γ -Fe and other austenitic

Case Study: Wang (2)

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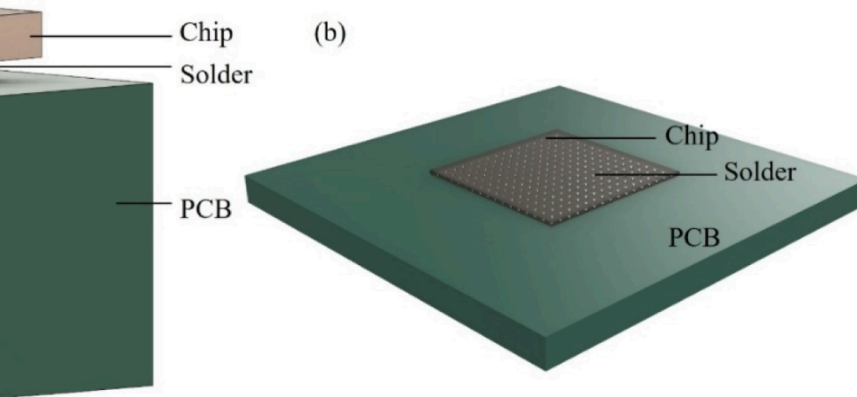


Source: Wang, C. H. (2001). "A Unified Creep–Plasticity Model for Solder Alloys." DOI: 10.1115/1.1371781

001) Apply to Solder

Why Wang's Paper Matters

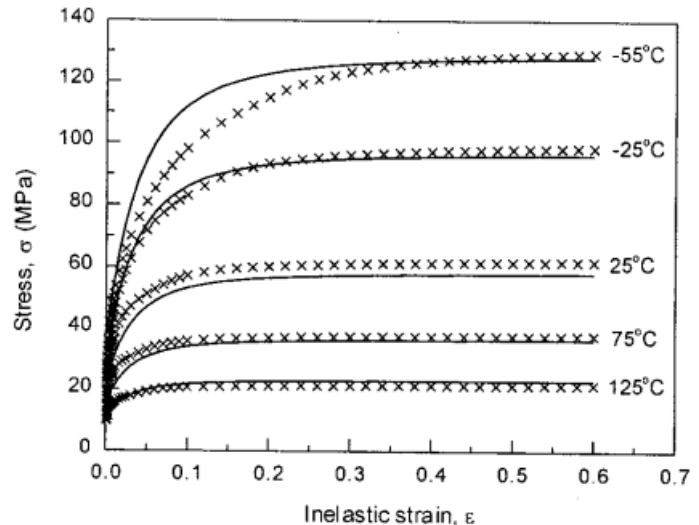
Wang's unified viscoplastic framework to model solder behavior.
The model can be reduced and fitted from experiments.
The theory into engineering-scale implementation.
Solder joints in microelectronic packages (chip on PCB, soldered
connections).



Observed Behavior

- **Top Graph (a):** $\dot{\epsilon} = 10^{-2} \text{ s}^{-1}$
- High strain rate $\rightarrow$ higher stress
- Recovery negligible $\rightarrow$ pronounced hardening
- **Bottom Graph (b):** $\dot{\epsilon} = 10^{-4} \text{ s}^{-1}$
- Lower strain rate $\rightarrow$ lower stress at same strain
- Recovery and creep effects more significant

Model Accuracy: Lines = model prediction, X = experimental data

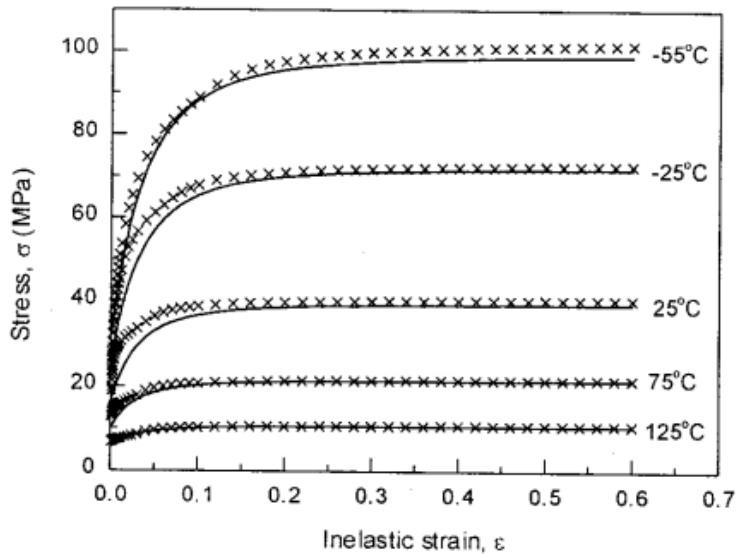


(a) $\dot{\epsilon} = 1.0 \times 10^{-2} \text{ s}^{-1}$

Key Insights from Wang (2001)

- “At lower strain rates, recovery dominates... the stress levels off early.”
- “At high strain rates, hardening dominates, and the stress grows continuously.”

Anand's model smoothly captures strain-rate and temperature dependence of solder materials.



(b) $\dot{\epsilon} = 1.0 \times 10^{-4} \text{ s}^{-1}$

Flow Rule (Plastic Strain Rate)

- $$\dot{\epsilon}^p = A \exp\left(-\frac{Q}{RT}\right) \left[\sinh\left(\frac{j\sigma}{s}\right) \right]^{1/m}$$
- Plastic strain rate increases with stress and temperature.
- No explicit yield surface; flow occurs at all nonzero stresses.

Deformation Resistance Saturation s^*

- $$s^* = \hat{s} \left(\frac{\dot{\epsilon}^p}{A} \exp\left(\frac{Q}{RT}\right) \right)^n$$
- Defines the steady-state value that s evolves toward.
- Depends on strain rate and temperature.

Evolution of Deformation Resistance s

- $$\dot{s} = h_0 \left| 1 - \frac{s}{s^*} \right|^a \text{sign} \left(1 - \frac{s}{s^*} \right) \dot{\epsilon}^p$$
- Describes dynamic hardening and softening of the material.
- s evolves depending on proximity to s^* and flow activity.

Note: Constants $A, Q, m, j, h_0, \hat{s}, n, a$ are material-specific and fitted to experimental creep/strain rate data.

Image Reference

Values are from correspond to 60Sn40Pb solder parameters used in Anand's model:

- S_0 : Initial deformation resistance
- Q/R : Activation energy over gas constant
- A : Pre-exponential factor for flow rate
- ξ : Multiplier of stress inside sinh
- m : Strain rate sensitivity of stress
- h_0 : Hardening/softening constant
- $\hat{s}$: Coefficient for saturation stress
- n : Strain rate sensitivity of saturation
- a : Strain rate sensitivity of hardening or softening

Numerical Values

- $S_0 = 5.633 \times 10^7 \text{ Pa}$
- $Q/R = 10830 \text{ K}$
- $A = 1.49 \times 10^7 \text{ s}^{-1}$
- $\xi = 11$
- $m = 0.303$
- $h_0 = 2.6408 \times 10^9 \text{ Pa}$
- $\hat{s} = 8.042 \times 10^7 \text{ Pa}$
- $n = 0.0231$
- $a = 1.34$

These constants match Wang's paper for modeling 60Sn40Pb viscoplasticity.

Initialization

- Material constants: $A, Q/R, j, m, h_0, \hat{s}, n, a, E$
- Strain rate: $\dot{\varepsilon}$
- Temperature set: $\{T_i\}$
- Set: $\varepsilon^p(0) = 0, \quad s(0) = \hat{s}$

Time Evolution Loop

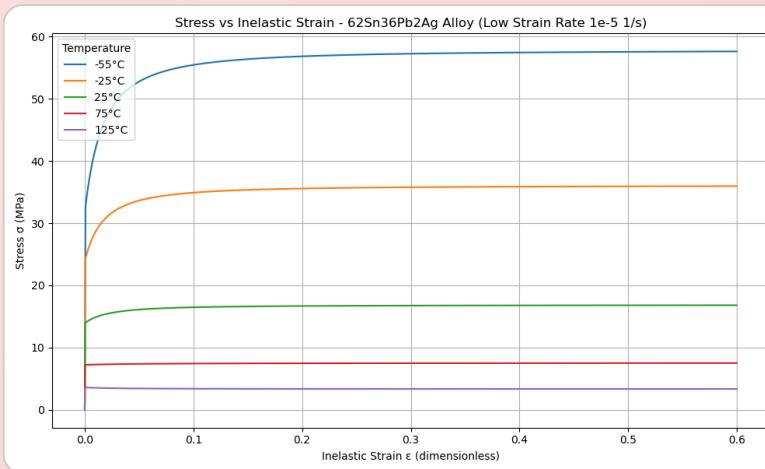
1. $\varepsilon_{\text{total}}(t) = \dot{\varepsilon} t$
2. $\sigma_{\text{trial}} = E(\varepsilon_{\text{total}} - \varepsilon^p)$
3. Compute $x = \frac{j\sigma}{s}$
4. Approximate $\sinh(x)$ (linearize if $|x| \ll 1$)
5. $\dot{\varepsilon}^p = Ae^{-Q/RT}(\sinh(x))^{1/m}$

Plastic Flow & Resistance Evolution

6. $s^* = \hat{s} \left(\frac{\dot{\varepsilon}^p}{A} e^{Q/RT} \right)^n$
7. $\dot{s} = h_0 \left| 1 - \frac{s}{s^*} \right|^a \text{sign} \left(1 - \frac{s}{s^*} \right) \dot{\varepsilon}^p$
8. Update: $\varepsilon^p(t + \Delta t) = \varepsilon^p(t) + \dot{\varepsilon}^p \Delta t$
9. Update: $s(t + \Delta t) = s(t) + \dot{s} \Delta t$
10. Record $(\varepsilon_{\text{total}}, \sigma_{\text{trial}})$

Termination

- Stop when $\varepsilon_{\text{total}} \geq \varepsilon_{\text{max}}$
- Plot σ vs ε for all T_i



```

import numpy as np
import matplotlib.pyplot as plt
from scipy.integrate import solve_ivp

# Material constants for 62Sn36Pb2Ag solder alloy
A = 2.24e8          # 1/s
Q_R = 11200        # K
j = 13             # dimensionless
m = 0.21           # dimensionless
h0 = 1.62e10       # Pa
s0 = 8.47e7        # Pa
s_hat = 8.47e7     # Pa
n = 0.0277         # dimensionless
a = 1.7            # dimensionless
E = 5.2e10         # Pa (Elastic modulus)

# Temperatures in Kelvin
T_C = [-55, -25, 25, 75, 125]
T_list = [T + 273.15 for T in T_C]

# Simulation parameters
strain_rate = 1e-5 # 1/s
eps_total_max = 0.6
t_max = eps_total_max / strain_rate
time_steps = 10000
t_eval = np.linspace(0, t_max, time_steps)

# Define the ODE system
def system(t, y, T):
    ep_p, s = y
    eps_total = strain_rate * t
    sigma_trial = E * (eps_total - ep_p)
    x = j * sigma_trial / s

    if np.abs(x) < 0.01:
        sinh_x = x
    else:
        sinh_x = np.sinh(np.clip(x, -30, 30))

    sinh_x = np.maximum(sinh_x, 1e-12)
    dep_p = A * np.exp(-Q_R / T) * sinh_x**(1/m)

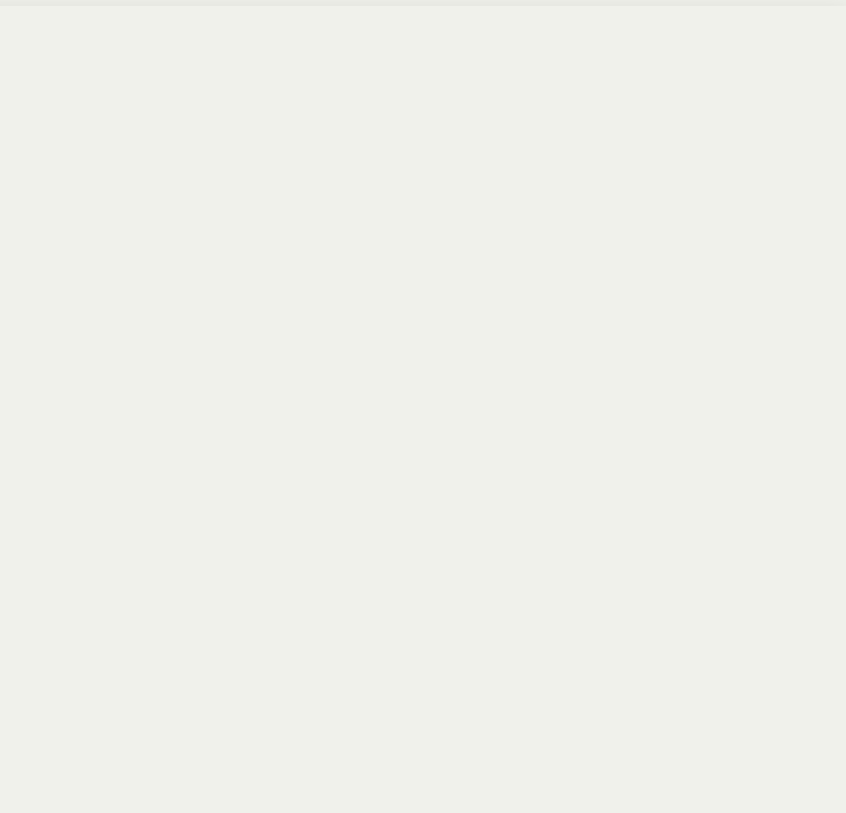
    s_star = s_hat * (dep_p / A * np.exp(Q_R / T))**n
    ds = h0 * np.abs(1 - s/s_star)**a * np.sign(1 - s/s_star) * dep_p

    return [dep_p, ds]

# Plotting

```

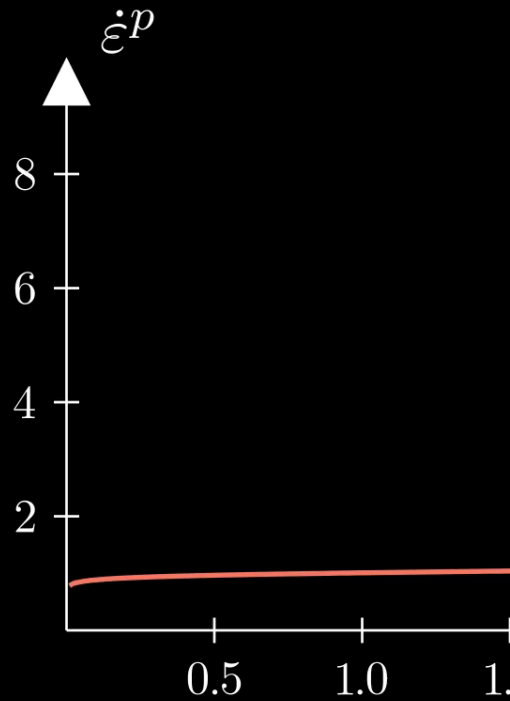
me for Anand Model



Strain rate sensi

- As $m \rightarrow 0$, rate insensitive
- As $m \rightarrow 1$, small stress ch

Anand Flow Law



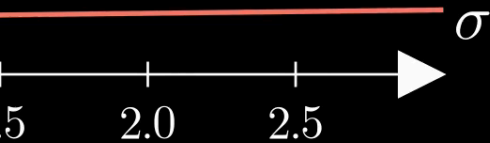
ivity of stress m

(yield)

ange causes big change in strain rate

ow: Varying m

$$m = 0.05$$



Tensorial Flow Rule (directional form)

$$\mathbf{D}^p = \dot{\epsilon}^p \left(\frac{3}{2} \frac{\mathbf{T}'}{\bar{\sigma}} \right)$$

Equivalent Stress Definition

$$\bar{\sigma} = \sqrt{\frac{3}{2} \mathbf{T}' : \mathbf{T}'}$$

Summary:

- Direction given by $\mathbf{T}'$.
- Magnitude determined by hyperbolic
- $\bar{\tau}$ represents the effective shear stress
- $\bar{\sigma} = \sqrt{\frac{3}{2} \mathbf{T}' : \mathbf{T}'}$ is the von Mises Eq
- Full flow = **direction** × **magnitude**.

Plastic Strain Rate (magnitude form)

$$\dot{\epsilon}^p = A \exp\left(-\frac{Q}{R\theta}\right) \left[\sinh\left(\xi \frac{\bar{\sigma}}{s}\right) \right]^{1/m}$$

Full Flow Rule with Hyperbolic Sine

$$\mathbf{D}^p = A \exp\left(-\frac{Q}{R\theta}\right) \left[\sinh\left(\xi \frac{\bar{\sigma}}{s}\right) \right]^{1/m} \left(\frac{3}{2} \frac{\mathbf{T}'}{\bar{\sigma}} \right),$$

$$= \dot{\gamma}^p \left(\frac{\tilde{\mathbf{T}}'}{2\bar{\tau}} \right), \quad \bar{\tau} = \left\{ \frac{1}{2} \text{tr}(\tilde{\mathbf{T}}'^2) \right\}^{1/2}$$

c sine based on $\bar{\sigma}/s$.

ss computed from deviatoric stress.

ivalent stress, but is formally defined without yield point

Stress Evolution Equation (Rate form of Hooke's Law)

$$\overset{\nabla}{\mathbf{T}} = \mathbb{L} [\mathbf{D} - \mathbf{D}^p] - \boldsymbol{\Pi} \dot{\theta}$$

(rate-form Hooke's law for finite deformation plasticity, with frame-indifference enforced through the Jaumann rate.)

Jaumann Rate Definition

$$\overset{\nabla}{\mathbf{T}} = \dot{\mathbf{T}} - \mathbf{W}\mathbf{T} + \mathbf{T}\mathbf{W}$$

Summary:

- Stress rate follows Jaumann rate
- Elastic response governs plasticity
- Thermal expansion introduces volumetric effects

Material Tensors and Operators

- $\mathbb{L} = 2\mu\mathbf{I} + (\kappa - \frac{2}{3}\mu)\mathbf{1} \otimes \mathbf{1}$ — isotropic elasticity tensor
- $\mathbb{L}\mathbf{D}$ represents how instantaneous strain rates generate stresses according to the elastic material's stiffness properties.
- $\mu = \mu(\theta)$, $\kappa = \kappa(\theta)$ — temperature-dependent moduli
- $\mathbf{\Pi} = (3\alpha\kappa)\mathbf{1}$ — stress-temperature coupling
- $\alpha = \alpha(\theta)$ — thermal expansion coefficient
- $\mathbf{D} = \text{sym}(\nabla \mathbf{v})$ — stretching tensor
- $\mathbf{W} = \text{skew}(\nabla \mathbf{v})$ — spin tensor
- $\mathbf{I}$ = fourth-order identity tensor
- $\mathbf{1}$ = second-order identity tensor

mann derivative to ensure frame indifference.
 ed by isotropic fourth-order tensor $\mathbb{L}$.
 duces additional stress through $\mathbf{\Pi}\dot{\theta}$.

Stress Evolution and Thermal Effects

In the stress evolution equation,

$$\dot{\mathbf{T}} = \mathbb{L} [\mathbf{D} - \mathbf{D}^p] - \mathbf{\Pi} \dot{\theta},$$

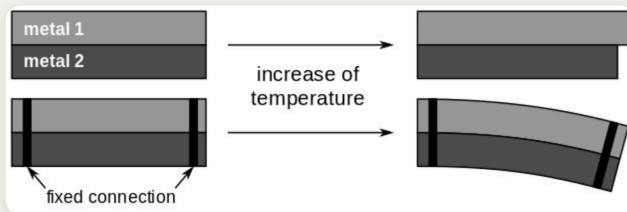
the term $\mathbf{\Pi} \dot{\theta}$ represents the stress change that would occur due to pure thermal expansion alone, without any mechanical loading.

Summary:

- Thermal expansion induces stress
- Subtracting $\mathbf{\Pi} \dot{\theta}$ ensures only mechanical loading
- This keeps the constitutive model

Why Subtract the Thermal Term?

- Thermal expansion creates strain even without external forces.
- Without subtracting $\mathbf{\Pi}\dot{\theta}$, the model would falsely attribute thermal strain as mechanical stress.
- Subtracting isolates the true mechanical response from thermal effects.



strain without force.
mechanical strains generate stresses.
model physically accurate during heating and cooling.

Context for the Relaxed Configuration

- The relaxed configuration represents the material after removing plastic deformations but before applying new elastic deformations.
- It is introduced to separate permanent plastic effects from recoverable elastic effects.
- All thermodynamic potentials, internal variables, and evolution laws are defined relative to this frame.
- The relaxed state provides a clean, natural reference for measuring elastic strain E^e and computing dissipation.

Sum

- The relaxed configuration isolates elastic responses from plastic evolution laws.

iate) Configuration

What Happens in the Relaxed Configuration?

- The elastic deformation gradient F^e is measured from the relaxed state to the current deformed state.
- Elastic strain measures like C^e and E^e are defined in this configuration.
- The Kirchhoff stress $\tilde{\mathbf{T}}$ is naturally associated with the relaxed volume.
- Plastic flow is accounted for separately through the plastic velocity gradient $\mathbf{L}^p$.

mary:

cleanly, enabling proper definition of thermodynamics and

Kinematics in the Relaxed Configuration

- Elastic deformation gradient:

$$F = F^e F^p \quad \Rightarrow \quad F^e = F F^{p-1}$$

- Elastic right Cauchy-Green tensor:

$$C^e = F^{eT} F^e$$

- Elastic Green–Lagrange strain tensor:

$$E^e = \frac{1}{2}(C^e - I)$$

Summary

- Elastic kinematics and stress measures are formulated in terms of plastic and elastic contributions.
- Stress Power Split allows Anand to cleanly isolate plastic and elastic contributions.
- Green-Lagrange strain tensor E^e is used because it is defined in the relaxed configuration
- The right Cauchy-Green tensor $C^e = F^{eT} F^e$ is required to compute the elastic deformation gradient F^e without referencing spatial coordinates.

Stress and Power Quantities

- Kirchhoff stress (weighted Cauchy stress):

$$\tilde{\mathbf{T}} = (\det F) \mathbf{T}$$

- Stress power split:

$$\dot{\omega} = \dot{\omega}^e + \dot{\omega}^p$$

$$\dot{\omega}^e = \tilde{\mathbf{T}} : \dot{\mathbf{E}}^e \quad , \quad \dot{\omega}^p = (C^e \tilde{\mathbf{T}}) : \mathbf{L}^p$$

Summary:

ed relative to the relaxed configuration, cleanly separating

lastic dissipation from elastic storage.

symmetrically captures nonlinear elastic strain relative to the

red as an intermediate to compute E^e from the elastic

coordinates

Thermodynamic Separation

1. Start with Total Dissipation:

$$\mathcal{D} = \dot{\omega} - \dot{\psi} \geq 0$$

where $\dot{\omega} = \hat{\mathbf{T}} : \dot{\mathbf{E}}^e + (\mathbf{C}^e \hat{\mathbf{T}}) : \mathbf{L}^p$

2. Split Stress Power:

$$\dot{\omega} = \dot{\omega}^e + \dot{\omega}^p$$

with:

- $\dot{\omega}^e = \hat{\mathbf{T}} : \dot{\mathbf{E}}^e$
- $\dot{\omega}^p = (\mathbf{C}^e \hat{\mathbf{T}}) : \mathbf{L}^p$

3. Group Terms with $\dot{\psi}$:

$$(\dot{\omega}^e - \dot{\psi}) + \dot{\omega}^p \geq 0$$

4. Apply Elastic Energy Consistency:

$$\dot{\omega}^e - \dot{\psi} = 0 \quad \Rightarrow \quad \dot{\omega}^p \geq 0$$

Key Physical Insights

- **Elastic deformations** are recoverable and do not cause entropy production.
- **All dissipation** stems from the plastic flow: $\dot{\omega}^p$.
- **Plastic work** increases entropy and governs viscoplastic evolution.

Summary:

The stress power split ensures that the second law is satisfied by assigning dissipation solely to irreversible processes.

Framework in the Reference Configuration

- The free energy ψ is defined relative to the reference configuration.
- State variables like E^e , θ , $\bar{g}$, $\bar{\mathbf{B}}$, s are used as arguments of ψ .
- Stress is expressed using the second Piola–Kirchhoff tensor $\mathbf{S}$.
- Dissipation inequality, stress–strain relations, and evolution laws are all written in reference variables.
- Mass density ρ_0 from the reference configuration normalizes all terms.

Summary

- In the reference configuration, all energy storage, stress, and evolution laws are written with reference-frame quantities for consistency and objectivity.

Configuration

Key Equations in the Reference Frame

- Free energy:

$$\psi = \psi(E^e, \theta, \bar{g}, \bar{\mathbf{B}}, s)$$

- Dissipation inequality:

$$\dot{\psi} + \eta \dot{\theta} - \rho_0^{-1} \mathbf{S} : \dot{\mathbf{E}} + (\rho_0 \theta)^{-1} \mathbf{q}_0 \cdot \mathbf{g}_0 \leq 0$$

- Constitutive relation:

$$\mathbf{S} = \rho_0 \frac{\partial \psi}{\partial E^e}$$

Summary:

Mass updates, and internal variable evolution are formulated
objectivity.

Thermodynamic Quantities

- Free energy density:

$$\boxed{\psi = \epsilon - \theta \eta}$$

- Reduced dissipation inequality:

$$\boxed{\dot{\psi} + \eta \dot{\theta} - \rho^{-1} \mathbf{T} : \mathbf{L} + (\rho \theta)^{-1} \mathbf{q} \cdot \mathbf{g} \leq 0}$$

- State variables:

$$\{E^e, \theta, \bar{g}, \bar{\mathbf{B}}, s\}$$

with E^e as elastic strain and s as internal resistance.

Summary:

- Free energy and dissipation
- Stress power naturally split
- Kirchhoff stress simplifies s

Stress Power and Kirchhoff Stress

- Stress power per relaxed volume:

$$\dot{\omega} = \left(\frac{\rho_0}{\rho} \right) \mathbf{T} : \mathbf{L}$$

- Weighted Cauchy (Kirchhoff) stress:

$$\tilde{\mathbf{T}} = (\det F) \mathbf{T} \quad \text{or} \quad \tilde{\mathbf{T}} = \left(\frac{\rho_0}{\rho} \right) \mathbf{T}$$

- Decomposition of stress power:

$$\dot{\omega} = \dot{\omega}^e + \dot{\omega}^p$$

$$\dot{\omega}^e = \tilde{\mathbf{T}} : \dot{\mathbf{E}}^e, \quad \dot{\omega}^p = (C^e \tilde{\mathbf{T}}) : \mathbf{L}^p$$

h govern thermodynamic consistency.
s into elastic and plastic parts.
stress evolution accounting for volume changes.

Multichamber J

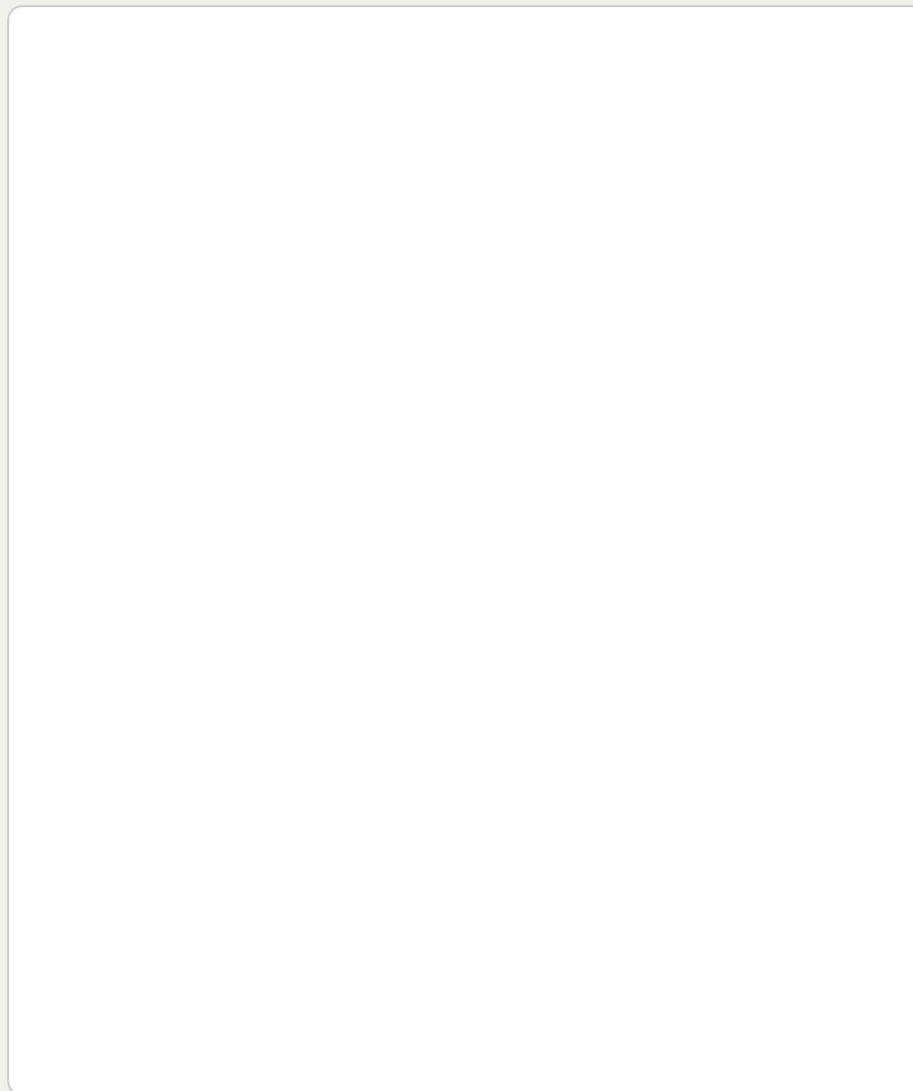
Michael Raba, MSc Candidate

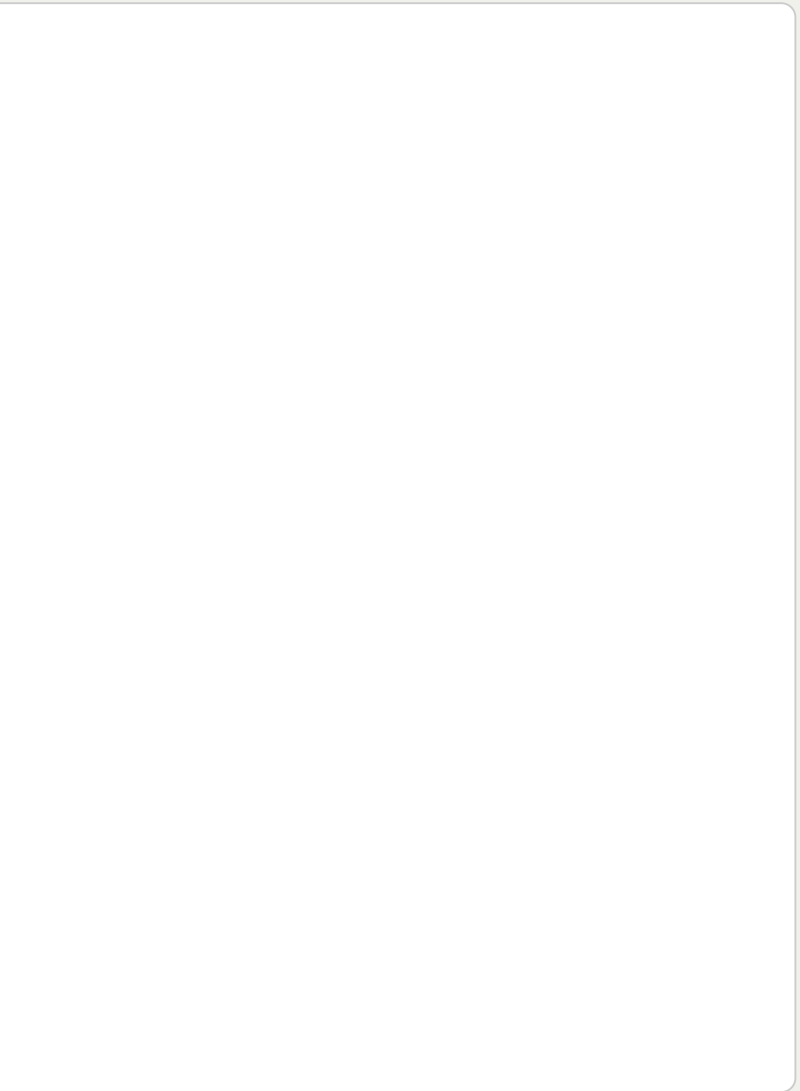
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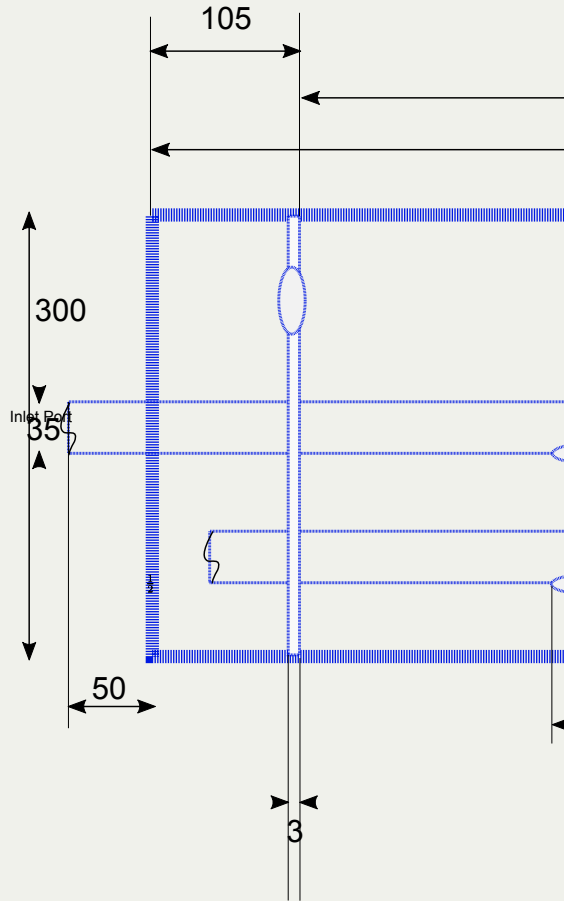
Muffler System

ate at University of Kentucky

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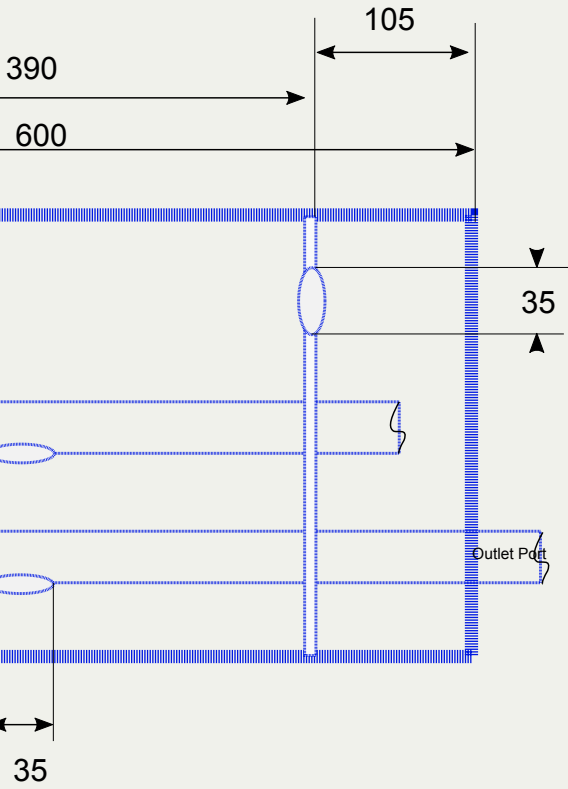


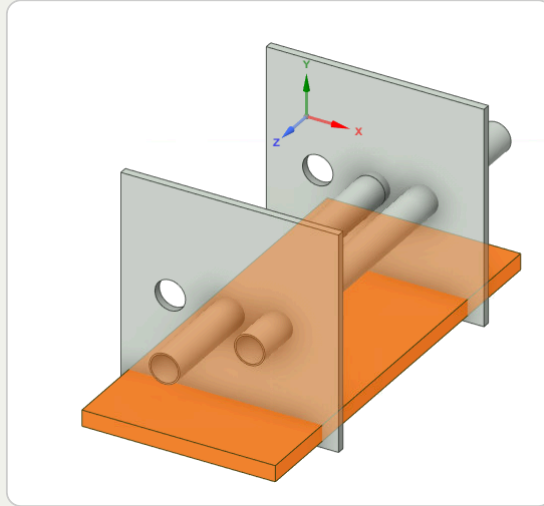




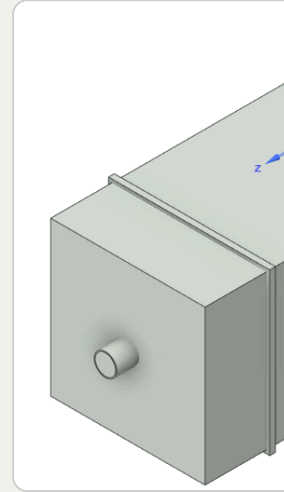
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al units in mm



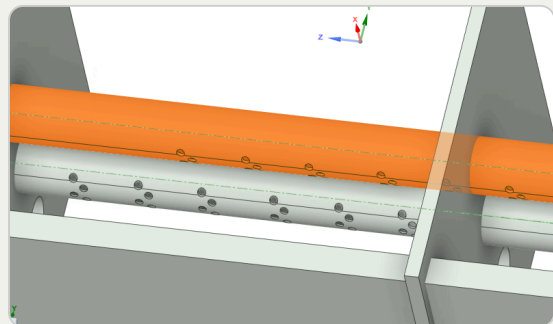


Part 1 — Chamber and Baffle

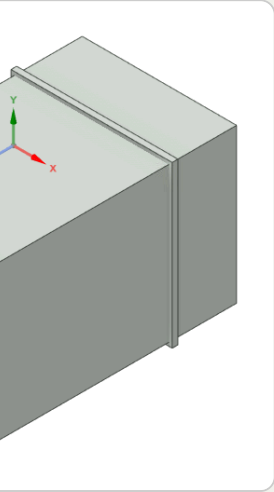


Part 2 — Front Panel

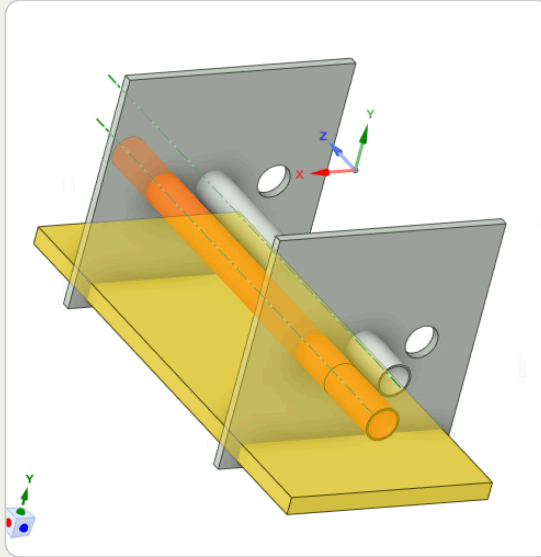
Part 4 — Showing perforates (aimed at fiberglass)



Muffler Subcomponents

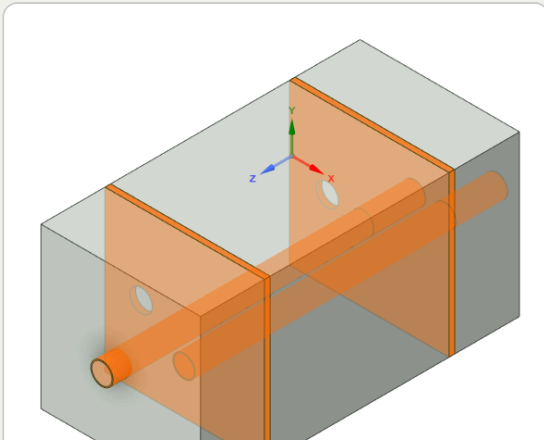


fluid domain



Part 3 — Fiberglass Absorbant (gold)

Part 5 — Final Assembly View



Simulated Transmission Loss (0–1000 Hz) by a

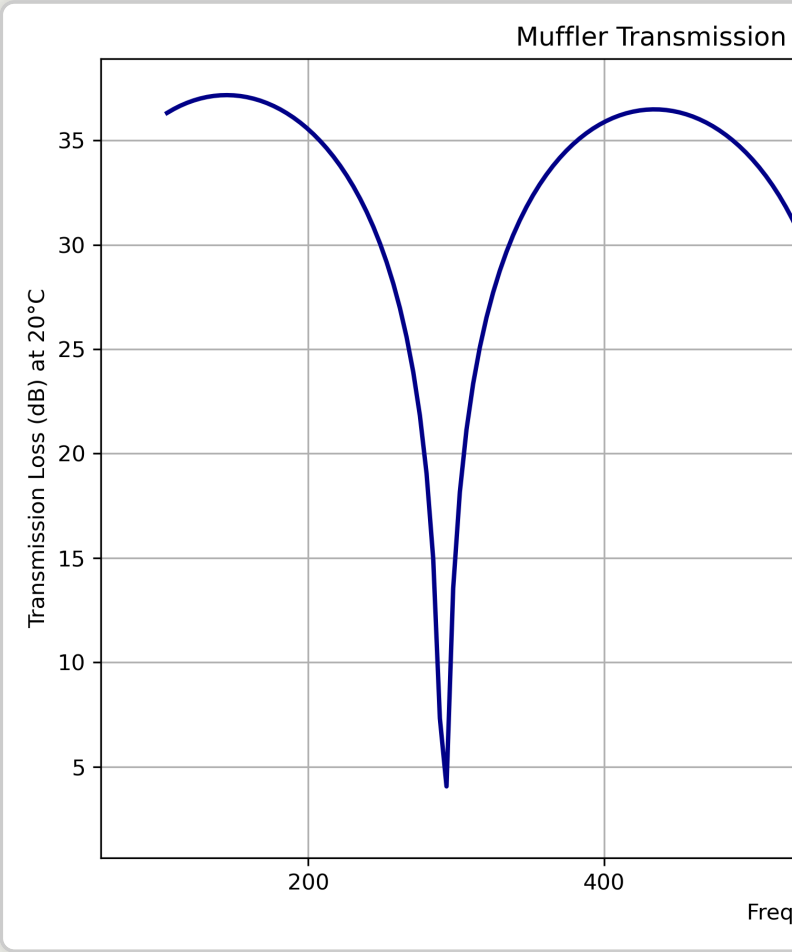
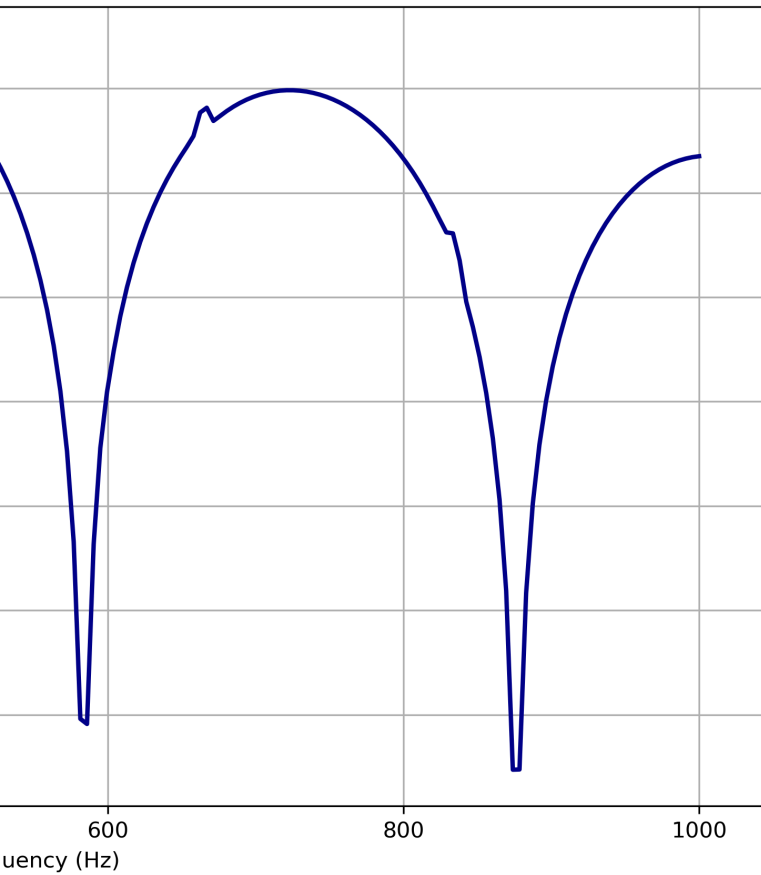


Figure: Transmission Loss curve of the m

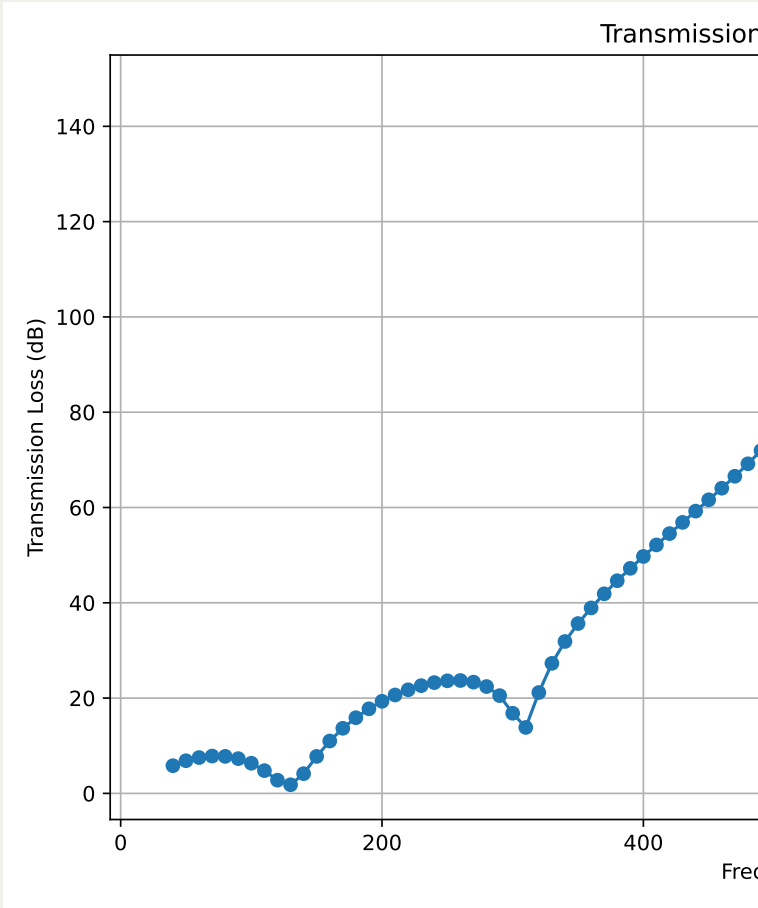
Simulation

approximating muffler walls as fluid at 20 deg C

Loss vs Frequency at 20°C



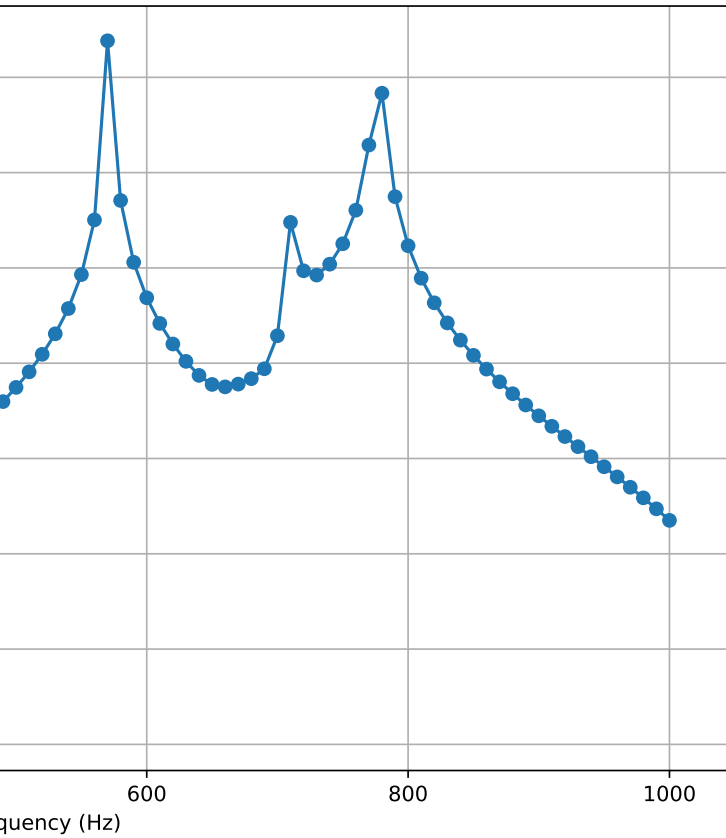
muffler between 5 Hz and 1000 Hz at 20°C.



Simulation

s (0–1000 Hz) Simlab model

n Loss vs Frequency



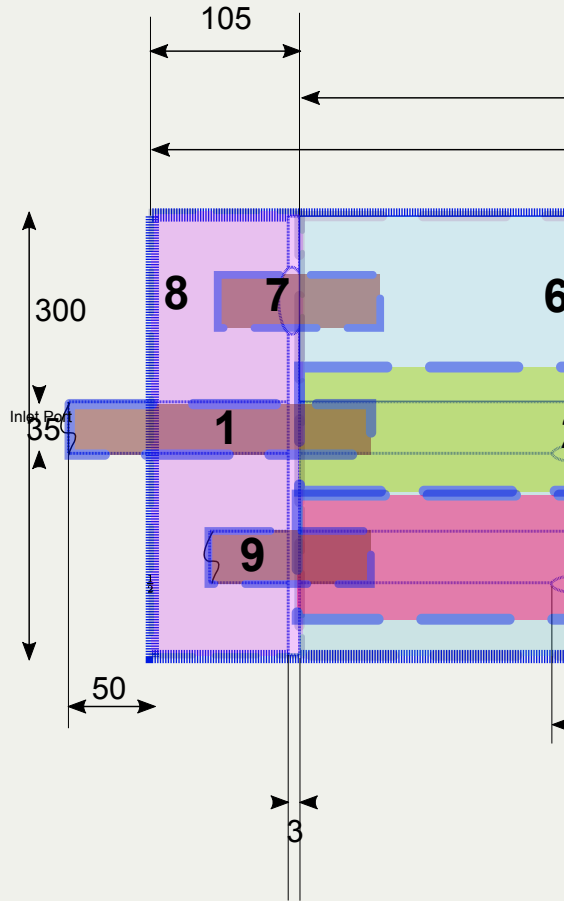
SIDLAB Model

- **File:** Mark3Sid.zip
- **Created with:** SIDLAB 5.1
- [↓ Download SIDLAB File](#)

ANSYS Simulation

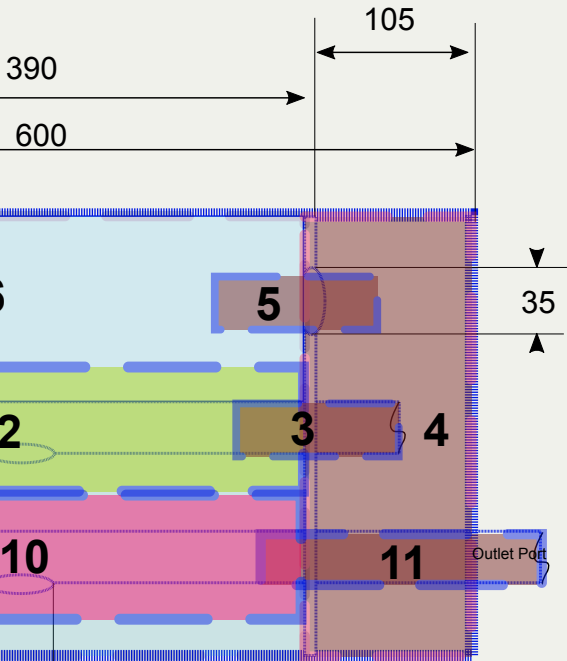
- **File:** Mark-I-MDF-cleaned-data.wbpz
- **Created with:** ANSYS 2023 R2
- [↓ Download ANSYS File](#)

dimension



Components

al units in mm



Sidlab Components

- | | |
|---------|----------|
| 1. Pipe | 7. Pipe |
| 2. QWT | 8. QWT |
| 3. Pipe | 9. Pipe |
| 4. QWT | 10. QWT |
| 5. Pipe | 11. Pipe |
| 6. QWT | |

Simulated vs Meas

Measured vs Simulated TL



Measured Insertion Loss



Insertion Loss Explanation

Insertion Loss (IL) quantifies how much sound is attenuated when a muffler is added to the system.

General formula:

$$IL = 10 \log_{10} \left(\frac{P_{\text{baseline}}}{P_{\text{muffler}}} \right)$$

Because our data is already in decibels (dB), this simplifies to:

$$IL = \text{Power}_{\text{baseline}} (\text{dB}) - \text{Power}_{\text{muffler}} (\text{dB})$$

1. Munjal ML. *Acoustics of Ducts and Mufflers*. 2nd ed. V
<https://doi.org/10.1002/9781118443125>
2. Dokumacı E. *Duct Acoustics: Fundamentals and Appli*
Press; 2021. ISBN: 9781108840750. <https://doi.org/10>

Note: These references are foundational texts in muffler and
schematic development, and

ences

Works

Wiley; 2014. ISBN: 9781118443125.

Applications to Mufflers and Silencers. Cambridge University
.1017/9781108840750

and duct acoustics and were consulted for system modeling,
and transmission loss analysis.

Update Slide

M. F.

Created: 2025-0

e for June 11

Raba

09-09 Tue 04:11

1. Code Execut

tion and Layout

1.1. L

1. b7.m
2. initSpectral.m
 - reads i
3. $\hookrightarrow$ initEigs.m
 - forms

layout

in binary files, takes eg m-fft

corrMat, finds eigenvalues

1. $\hookrightarrow$ initPod.m

- carries out POD calculations (quadrature, multiplication)
Hellstrom Smits 2017 for Snapshot POD)

2. $\hookrightarrow$ timeReconstructFlow.m

- performs 2d reconstruction + plotSkmr (generates 1d ra

Layout 2

ggf between $\alpha\Phi$) according to Papers (Citriniti George 2000 for Classic POD,

dial graph)

1.3. Importa

`pipe = Pipe();` creates a Pipe Class. As the function

1. `obj.CaseId` - stores properties like Re , rotation number S , experiment
frequently called vectors (`rMat` $r = 1, \dots, 0.5$)
2. `obj.pod` - eigen data, used for calculating POD
3. `obj.solution` - computed POD modes
4. `obj.plt` - plot configuration

ant Switches

ns (above) are called, data is stored in sub-structs:

al flags such as quadrature (simpson/trapezoidal), number of gridpoints,

2. Equations Used

in Code Procedure

2.1. Classic P

The following equations a

$$\int_{r'} \mathbf{S}(k; m; r, r') \Phi^{(n)}(k; m; r')$$

$$\mathbf{S}(k; m; r, r') = \lim_{\tau \rightarrow \infty} \frac{1}{\tau} \int_0^\tau \mathbf{u}(k$$

$$\alpha^{(n)}(k; m; t) = \int_r \mathbf{u}(k; m; r, t) \Phi$$

OD Equations

are used in the above code.

$$r' \mathrm{d}r' = \lambda^{(n)}(k; m) \Phi^{(n)}(k; m; r)$$

$$; m; r, t) \mathbf{u}^*(k; m; r', t) \mathrm{d}t$$

$$\Phi^{(n)*}(k; m; r) r \mathrm{d}r$$

2.2. Classic POD

$$\begin{aligned} &\int_{r'} \underbrace{r^{1/2} S_{i,j}(r,r';m;f) r^{1/2}}_{W_{i,j}(r,r';m;f)} \\ &= \underbrace{\lambda^{(n)}(m,f)}_{\hat{\lambda}^{(n)}(m;f)} \underbrace{r^{1/2} \phi_i^{(n)}(r;)}_{\hat{\phi}_i^{(n)}(r,m;f)} \end{aligned}$$

$$\alpha_n(m;t) = \int_r \mathbf{u}(m;r,t) r$$

Equations (Fixed)

$$\underbrace{\int_0^{\infty} \phi_j^{*(n)}(r';m;f) \, r'^{1/2} \, \mathrm{d}r'}_{\hat{\phi}_j^{\psi(i)}(r';m;f)} \\
 \underbrace{\phantom{\int_0^{\infty} \phi_j^{*(n)}(r';m;f) \, r'^{1/2} \, \mathrm{d}r'}}_{m;f)} \\
 f) \\
 \int_0^{\infty} r^{1/2} \Phi_n^*(m;r) \, \mathrm{d}r$$

2.3. Snapshot R

$$\begin{aligned}
& \lim_{\tau \rightarrow \infty} \frac{1}{\tau} \int_0^\tau \mathbf{u}_T(k; m; r, t) \\
& = \Phi_T^{(n)}(k; m; r) \lambda^{(n)}(k; m) \\
\mathbf{R}(k; m; t, t') &= \int_r \mathbf{u}(k; m; r, t) \\
& \lim_{\tau \rightarrow \infty} \frac{1}{\tau} \int_0^\tau \mathbf{u}_T(k; m; r, t) \\
& = \Phi_T^{(n)}(k; m; r) \lambda^{(n)}(k; m)
\end{aligned}$$

POD Equations

$$\alpha^{(n)*}(k;m;t)dt$$

)

$$n;r,t)\mathbf{u}^*(k;m;r,t')r\,dr$$

$$\alpha^{(n)*}(k;m;t)\,dt$$

).

2.4. Recon

The reconstruc

$$q(\xi, t) - \bar{q}(\xi) \approx \sum_{j=1}^r a_j(t) \varphi_j(\xi)$$

$$q(r, \theta, t; x) = \bar{q}(r, \theta, t; x) +$$

Since the snapshot pod implementation is not error-free, the re

$$q(r, \theta, t; x) = \bar{q}(r, \theta, t; x) + (\text{factor}$$

construction

tion is given by

$\Rightarrow$

$$\sum_{n=1} \sum_{m=0} \alpha^{(n)}(m; t) \Phi^{(n)}(r; m; x)$$

construction can only be recovered by writing for factor $\gg 0$.

$$r \gamma) \sum_{n=1} \sum_{m=0} \alpha^{(n)}(m; t) \Phi^{(n)}(r; m; x)$$

2.5. Reco

In order to reconstruct in code, `caseId.fluctuation = 'off'`. The

nstruction

this is incorrect. The necessary use of (factor γ) is incorrect

3. Der

To derive the questioned eq

$$\frac{1}{\tau} \int_0^\tau \mathbf{u}_T(k; m; r,$$

Substitute $\mathbf{u}_T$ w

$$\frac{1}{\tau} \int_0^\tau \left(\sum_l \Phi_T^{(l)}(k; m; r) \alpha^{(l)} \right.$$

ivation

uation, consider the integral:

$$t) \alpha^{(n)*}(k; m; t) dt.$$

ith its expansion:

$$) \alpha^{(n)*}(k; m; t) \Big) \alpha^{(n)*}(k; m; t) dt.$$

Exchange the order of summation and

$$\sum_l \Phi_{\text{T}}^{(l)}(k; m; r) \left(\frac{1}{\tau} \int_0^\tau \alpha^{(l)} \right)$$

Due to the orthogonality, namely t

$$\langle a^{(n)} \alpha^{(p)} \rangle$$

all terms where $l \neq n$ will vanish, and

$$\Phi_{\text{T}}^{(n)}(k; m; r) \left(\frac{1}{\tau} \int_0^\tau \alpha^{(n)} \right)$$

This derivation assumes the normalization of modes and their orthogonality, form that reveals the spatial structure ($\Phi_{\text{T}}^{(n)}$)

derivation

and integration, and apply orthogonality,

$$\int_0^T \alpha^{(l)}(k; m; t) \alpha^{(n)*}(k; m; t) dt \Bigg).$$

that $\alpha^{(n)}$ and $\alpha^{(p)}$ are uncorrelated

$$= \lambda^{(n)} \delta_{np}$$

and there remains only the $l = n$ term,

$$\int_0^T \alpha^{(n)}(k; m; t) \alpha^{(n)*}(k; m; t) dt \Bigg).$$

along with the eigenvalue relationship to simplify the original integral into a sum of each mode scaled by its significance $(\lambda^{(n)})$.

The cross-correlation tensor $\mathbf{R}$ is defined as $\mathbf{R}(k; m; t, t') = \int_r \mathbf{u}(k; m; r, t) \mathbf{u}(k; m; r, t')$ tensor. The n POD m

$$\lim_{\tau \rightarrow \infty} \frac{1}{\tau} \int_0^\tau \mathbf{u}_T(k; m; r, t) \alpha^{(n)*}(k; r, t) dt$$

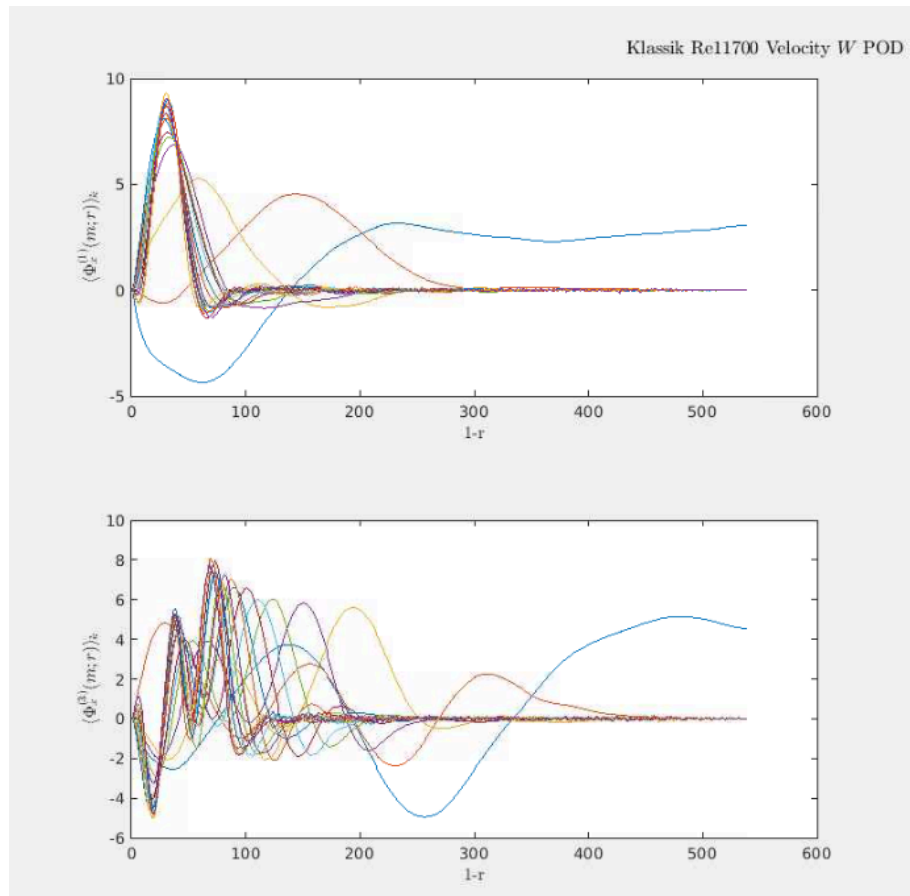
erivation

$t) \mathbf{u}^*(k; m; r, t') r \, \mathrm{d}r$. This tensor is now transformed from $[3r \times 3r']$ to a nodes are then constructed as,

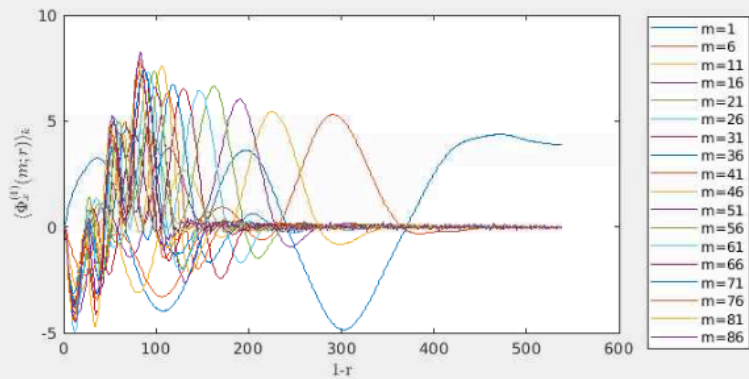
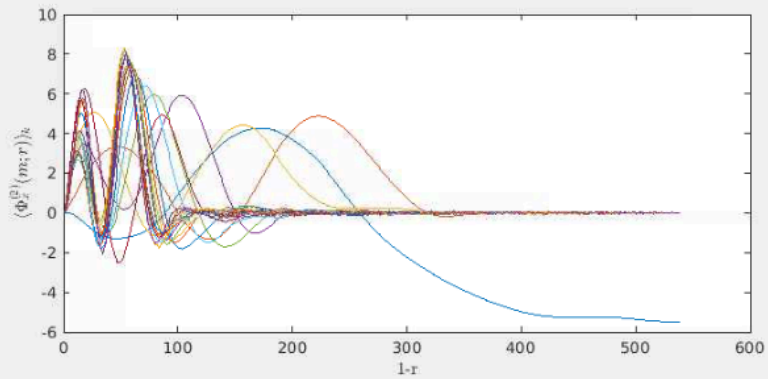
$$n; t) \mathrm{d}t = \Phi_{\mathrm{T}}^{(n)}(k; m; r) \lambda^{(n)}(k; m).$$

4. Result Comparison

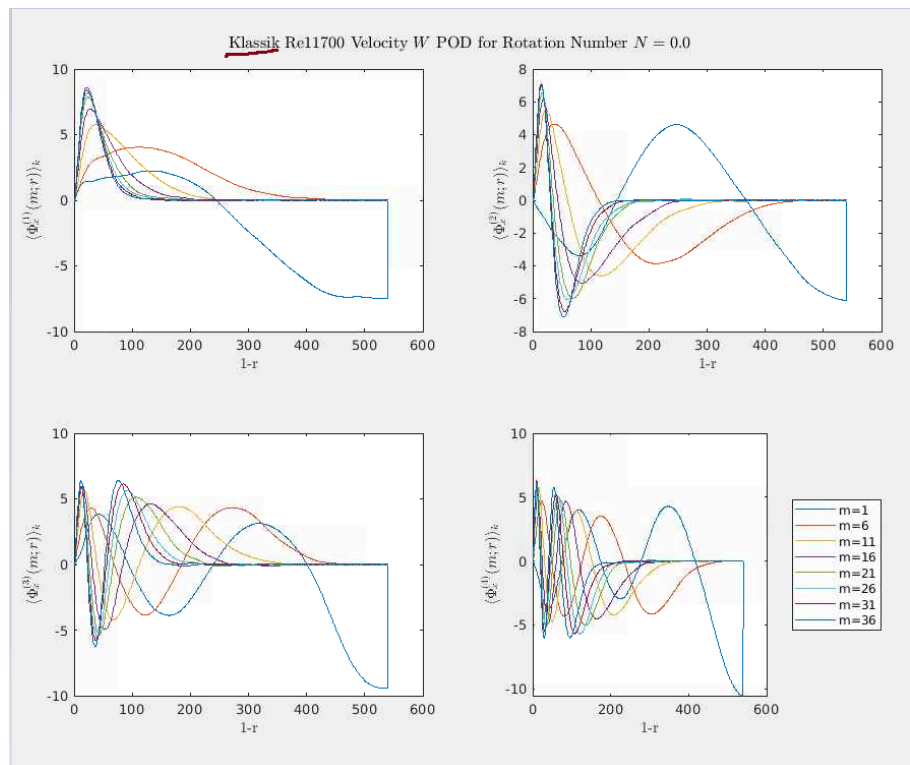
on Classic/Snapshot



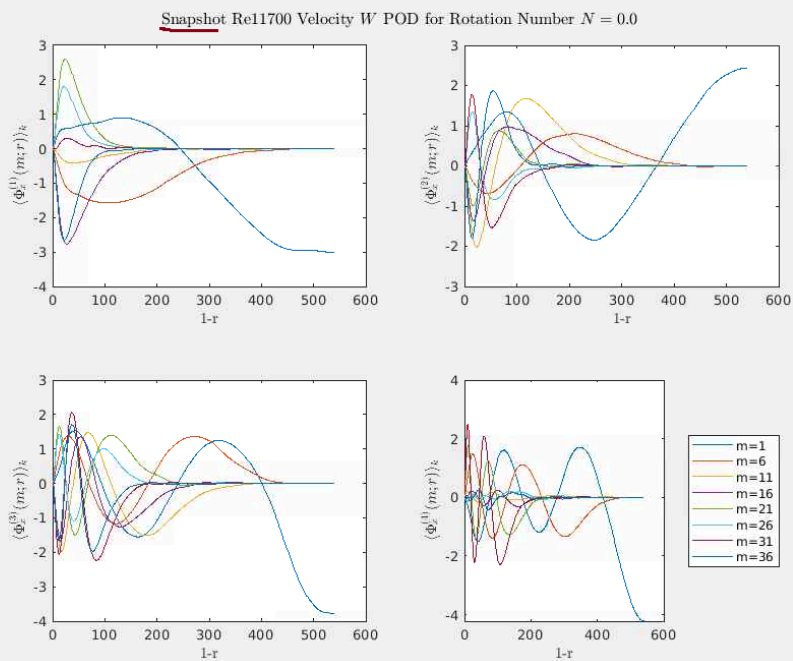
for Rotation Number $N = 3.0$

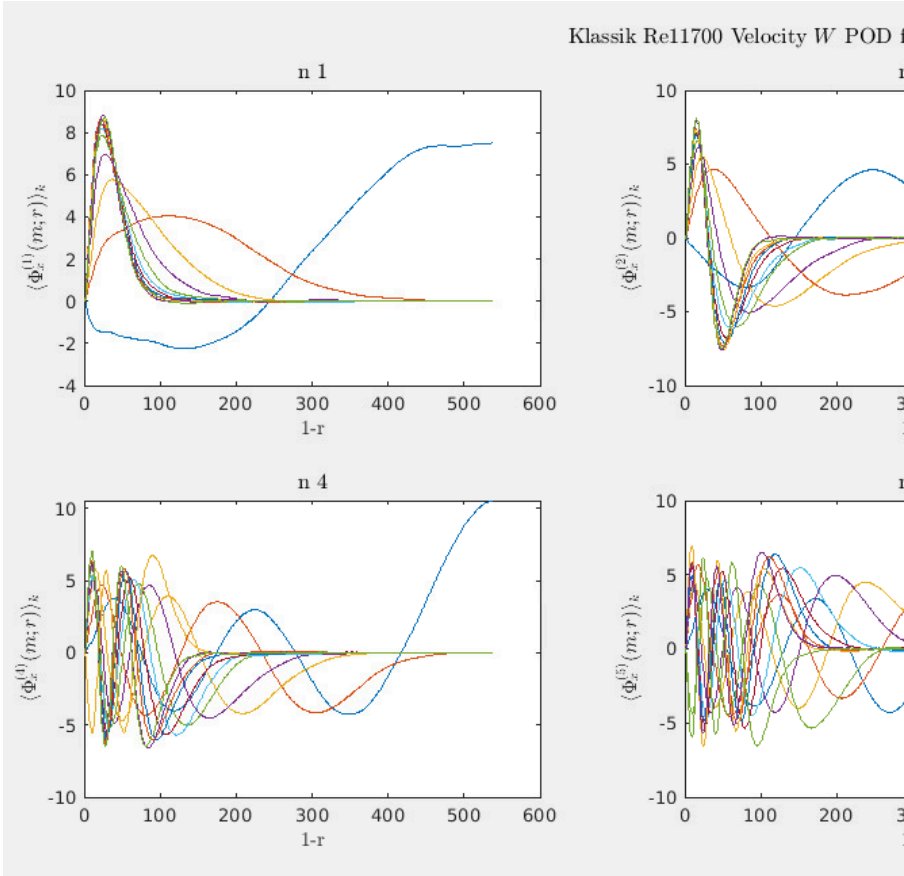


4.2. Snapshot-CL



Classic Comparison

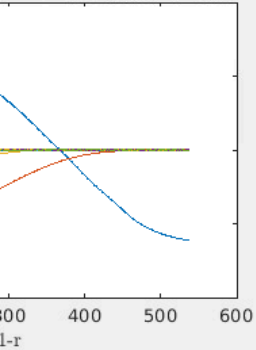




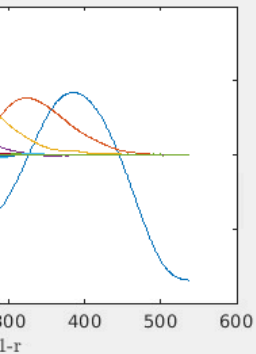
POD S=0.0

For Rotation Number $N = 0.0$

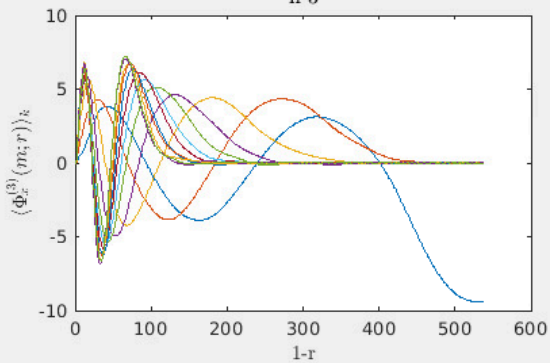
n 2



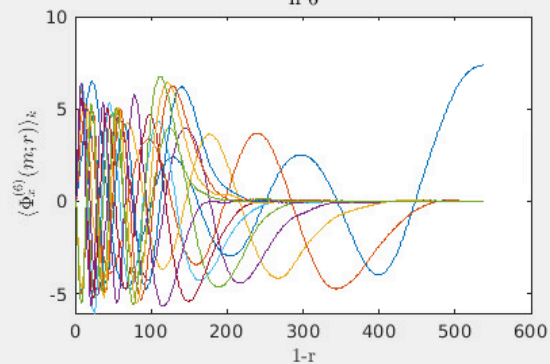
n 5

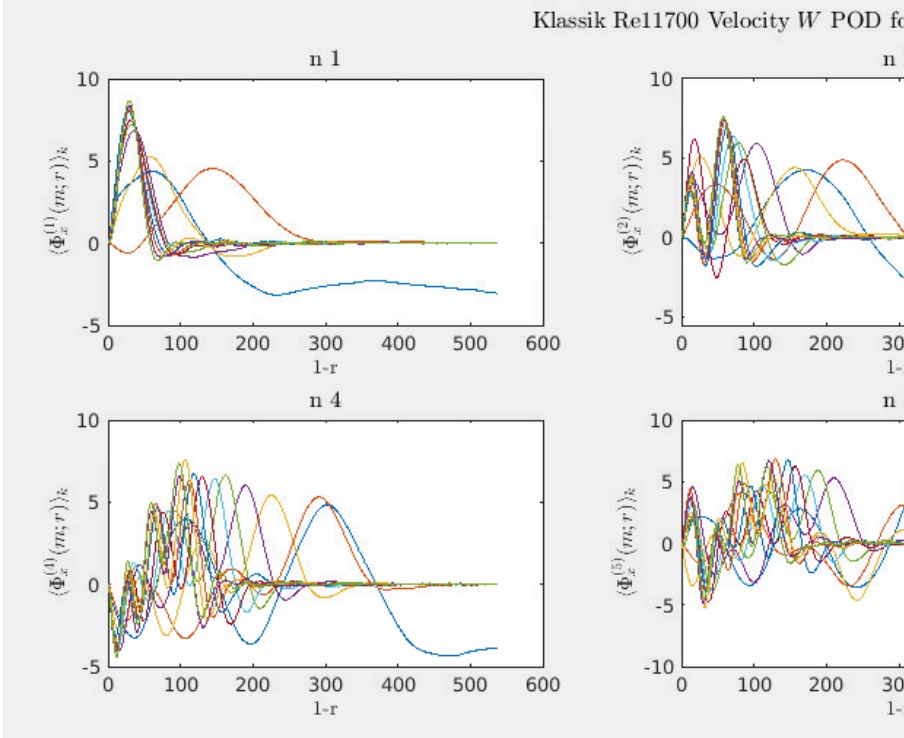


n 3



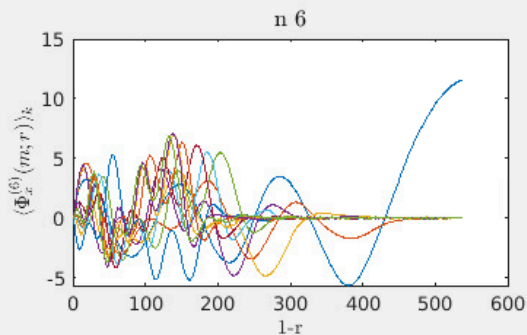
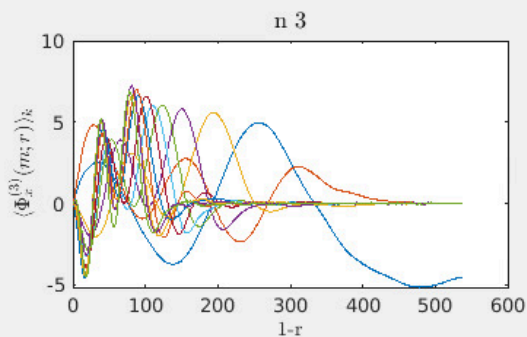
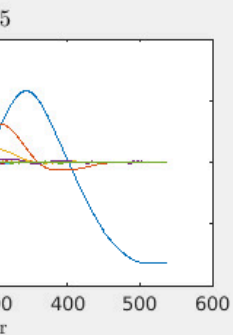
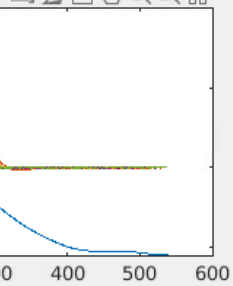
n 6

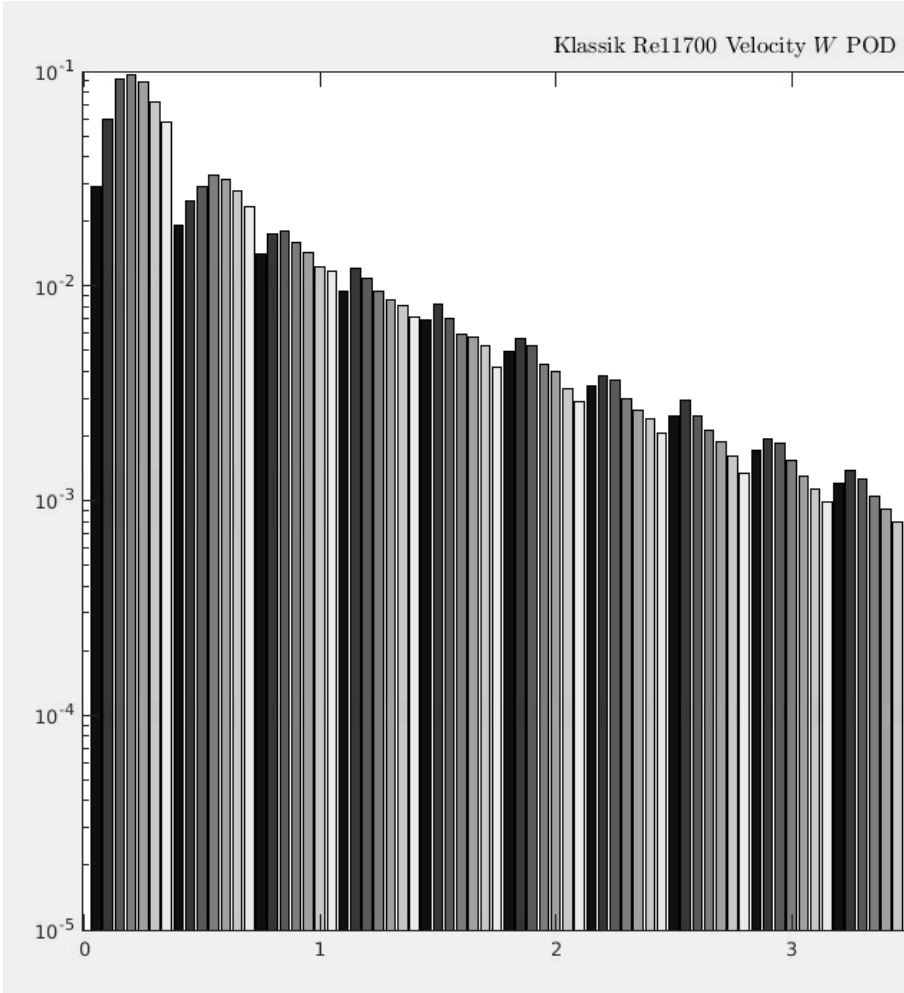




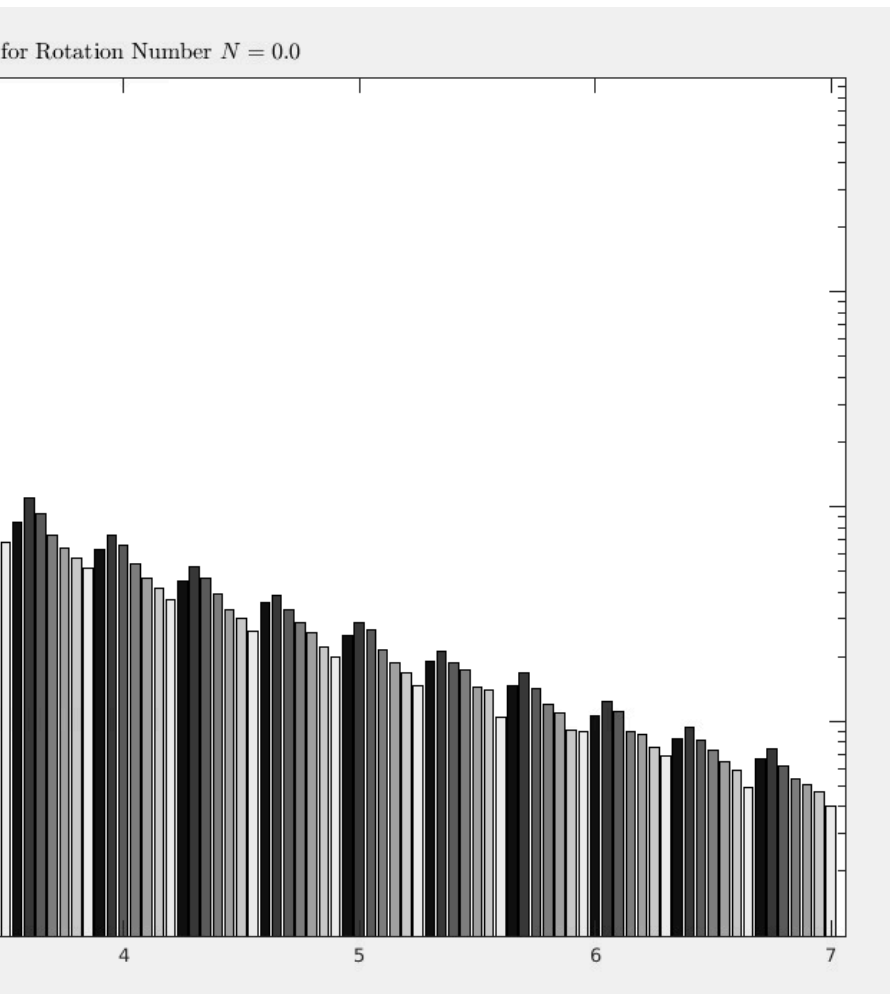
POD S=3.0

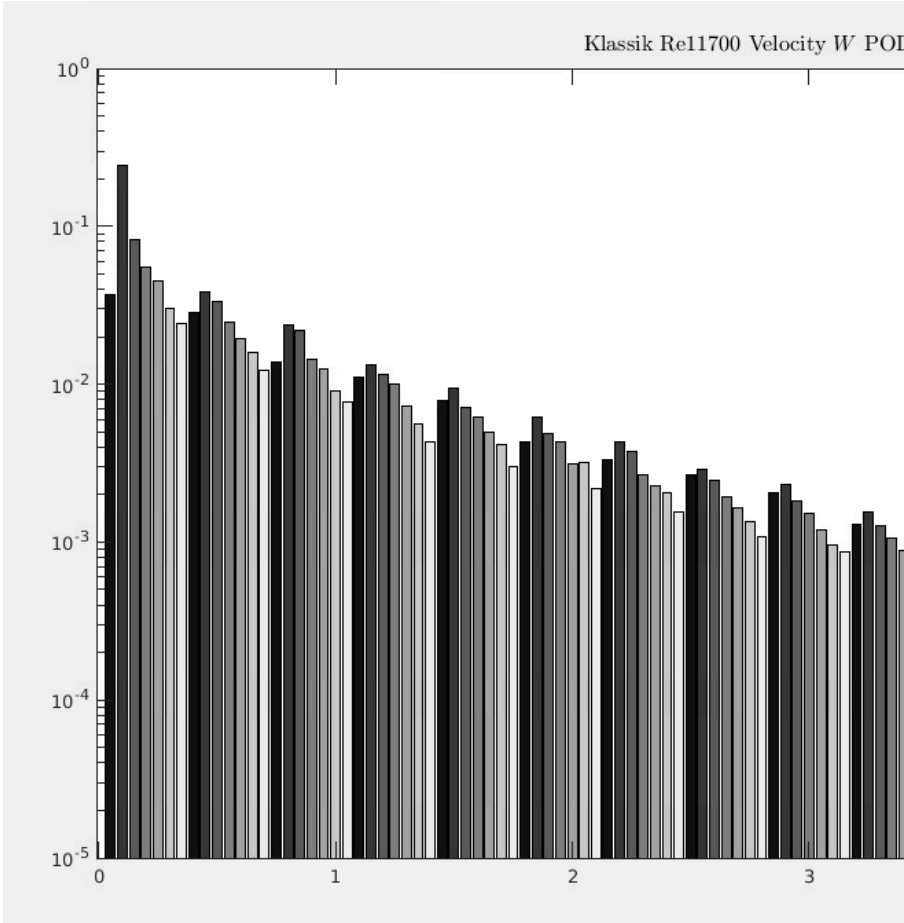
for Rotation Number $N = 3.0$



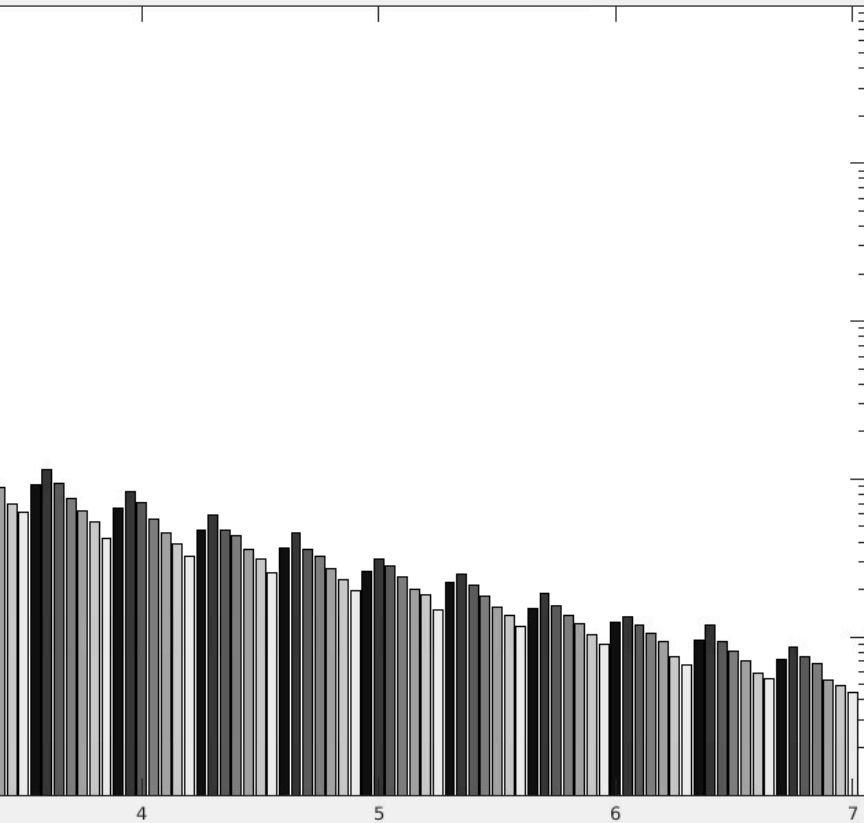


n=0 Classic





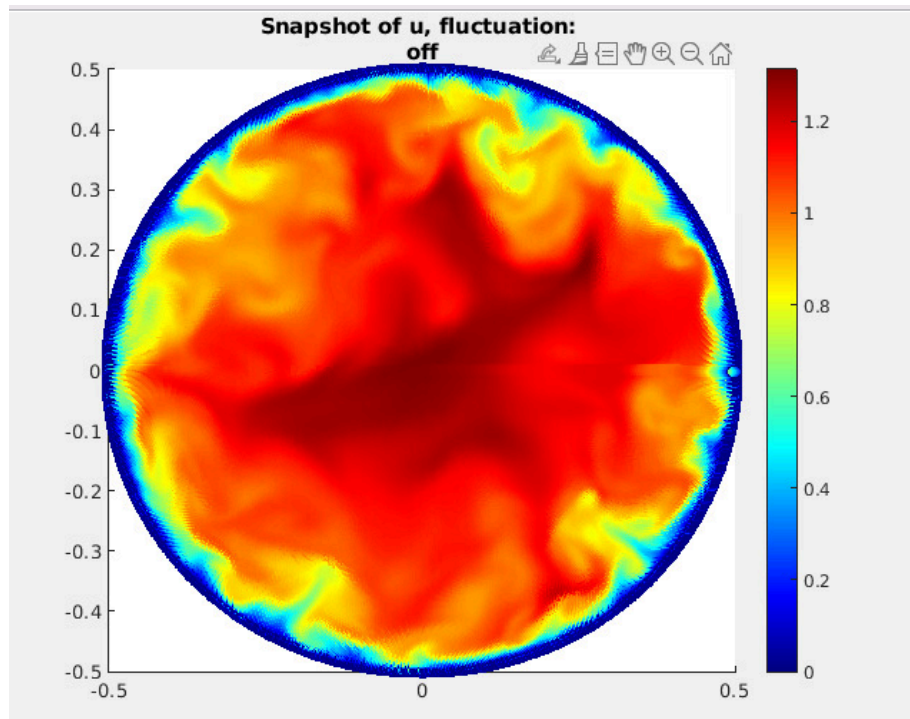
D for Rotation Number $N = 3.0$



analysis

6. Recon

struction



nstruction

